

Scalability & Future Plan

Date	30 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document describes the current limitations of the Human Development Index Prediction system and presents future enhancements to improve scalability, usability, performance, and security. The proposed roadmap aims to support larger datasets, more users, and additional intelligent features in future versions.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited input features for prediction	May slightly affect prediction accuracy	High
2	No user authentication or login system	Cannot store user prediction history	Medium
3	Limited country list in the user interface	Users can select only predefined countries	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent users	Deploy on scalable cloud infrastructure with load balancing
2	Data Storage	Model stored locally (.pkl)	Integrate cloud database and model storage
3	Performance	Suitable for small-scale usage	Optimize model and enable caching for faster predictions
4	Security	Basic Flask application	Implement HTTPS, authentication, and input validation

Future Roadmap:

The future development of this project focuses on improving prediction accuracy, expanding supported countries, enhancing security, and providing advanced visualization features. The application can also be integrated with cloud databases and real-time datasets to support continuous model updates and improved user experience.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Add more countries and input features	1–2 Months	Improve prediction accuracy
Phase 2	Integrate database and user login	2–3 Months	Store prediction history and user data
Phase 3	Deploy on scalable cloud platform with API support	3–4 Months	Improve scalability and accessibility
Phase 4	Interactive dashboards and real-time HDI analytics	4–6 Months	Better visualization and decision support